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Credit - Drishti Ias

Indian Coast Guard Inducts Indigenous ALH MK-III Helicopter

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Source: AIR

Recently, the **Indian Coast Guard** inducted the **indigenous Advanced Light Helicopter (ALH) MK-III,** designated **CG 875** , under **Coast Guard Region (West).**

- CG 875 is the **first of five allotted ALH MK-III helicopters** to be positioned with 831 Squadron (CG) at **Ratnagiri, Maharashtra.**

ALH MK-III

- **About:** Advanced Light Helicopter (ALH) MK-III is an **indigenous multi-role helicopter** manufactured by **Hindustan Aeronautics Limited (HAL).**
- **Advanced Equipment:** It features advanced radar, electro-optical sensors, Shakti engines, a full glass cockpit, a high-intensity searchlight, advanced communication systems, an **Automatic Identification System (AIS)** and a search-and-rescue homing system.
- **Operational Role** : It supports **maritime reconnaissance, surveillance and search-and-rescue operations** over extended ranges, operating from ships during both day and night.
- **Multi-role Capability:** The helicopter can switch from an offensive platform equipped with a heavy machine gun to a medical evacuation platform fitted with a medical intensive care unit for transporting critically ill patients.
- **Strategic Significance** : Its induction strengthens the Indian Coast Guard's maritime surveillance, threat detection, coastal security and emergency response capabilities, including the protection of India's **Exclusive Economic Zone (EEZ).**

Read more: **Advanced Light Helicopter (ALH) MK-III Squadron**

IMD Launches Interactive Cyclone Warning and Marine Weather Platform

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Source: DD

The **India Meteorological Department (IMD)** launched an interactive **Cyclone Warning and Marine Weather Services Decision Support System(DSS)** to strengthen cyclone preparedness and support timely decision-making.

- **About DSS:** It is a **WebGIS-based platform** providing interactive maps and visualisations of cyclone-related hazards, including cyclone tracks, intensity forecasts, wind distribution, rainfall warnings, storm surge information and rainfall forecasts for river catchments.
- **Background:** The launch coincided with the anniversaries of the landfalls of **Very Severe Cyclonic Storms Phailin (2013)** and Hudhud (2014), whose successful forecasting marked important milestones in India's cyclone early-warning capabilities.
- **Objective:** It provides **location-specific hazard information** to disaster management authorities, government agencies and the public to improve preparedness and reduce losses.
- **Accessibility** : The platform is accessible through IMD and Regional Specialised Meteorological Centre((RSMC) websites and IMD’s Decision Support System.
- **Institutional Coordination:** IMD's **biannual Pre-Cyclone Exercise Meeting** , conducted since 2008, brings together disaster management authorities, defence forces, coastal states, fisheries departments and scientific organisations **to review cyclone preparedness.**
- **Significance:** The platform enables location-specific early warnings, supports evacuation planning and **disaster risk reduction, and improves coordination** among forecasting agencies, disaster management authorities and other stakeholders.

Read more: India’s Cyclone Preparedness

NASA’s Perseverance Rover Sets a New Driving Record on Mars

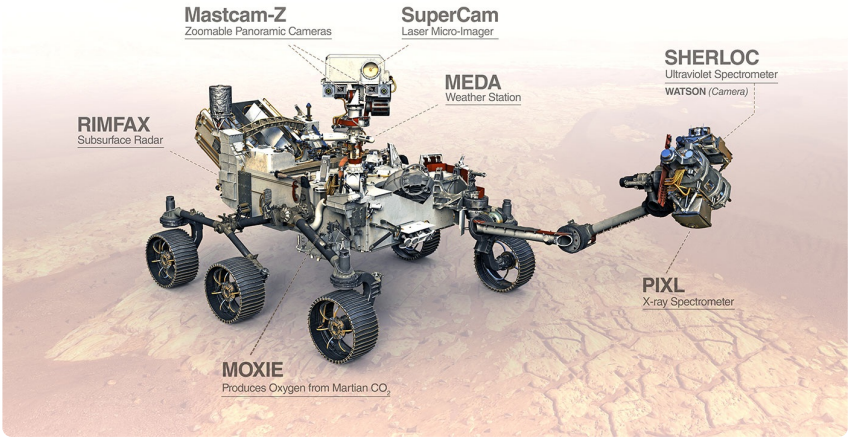
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Source: TP

NASA’s Perseverance rover has travelled 45.22 km on Mars, surpassing the previous off-Earth

driving record of **45.16 km held by NASA's Opportunity rover,** demonstrating the capabilities of autonomous navigation in planetary exploration.

- **About Perseverance:** It is a **car-sized, six-wheeled robotic rover** developed by NASA to investigate **Mars** ' geology, ancient habitability and potential evidence of past microbial life. It primarily explores **Jezero Crater** , a region believed to have contained an **ancient lake** and river delta billions of years ago.
- **Objective** : To investigate evidence of ancient microbial life on Mars, study its geology and evolution, collect rock and regolith samples for potential return to Earth, and identify environments that may have supported life.
- **Important Technologies:** Perseverance uses **AutoNav f** or autonomous navigation, **the Vision Compute Element** for rapid image processing, and **a plutonium-238-powered Multi-Mission Radioisotope Thermoelectric Generator (MMRTG) for electricity generation** .
- **Findings:** NASA has reported a **potential biosignature** in ancient rocks associated with a former river channel feeding Jezero Crater but this is not confirmation that life existed on Mars.
- **Extended Mission:** NASA has extended the mission to explore the crater rim until approximately the end of 2028, although the rover may remain operational beyond this period.
- **Opportunity Rover:** It was launched alongside its twin **Spirit** , landed on Mars in 2004 to investigate the planet's geology and past conditions suitable for microbial life. It found evidence of **ancient habitable conditions on Mars** and operated for nearly 15 years, exceeding its planned mission duration by about 60 times.



Read more: **Ancient Lake on Mars**

Indian Army Launches Bharat Rice

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Source: TH

The **Indian Army** launched ' **Bharat Rice'** for defence personnel and signed a five-year Memorandum of Understanding (MoU) with the **Food Corporation of India (FCI)** , to strengthen military logistics and operational preparedness.

- **About Bharat Rice:** It is a **superior-quality rice initiative** developed based on feedback from troops, with rice supplied according to prescribed **defence specifications to improve ration quality and logistical efficiency.**
- **Objective** : To improve ration quality, ensure timely replenishment and strengthen military preparedness, especially during emergencies and rapid mobilisation.
- **Supply Mechanism** : Rice will be delivered **directly** to Army supply depots, including those in remote areas of Ladakh and the Northeast, reducing the need for depots to collect supplies from local FCI facilities.
- **Role of NAFED:** The **National Agricultural Cooperative Marketing Federation of India (NAFED)** will supply the rice during the financial year 2026–27.
- **Packaging and Quality Control:** Rice will undergo quality checks and be supplied in 10-kg and 30-kg bags. Ergonomic packaging will facilitate handling and transportation in difficult terrain.
- **MoU Highlights:** The agreement gives the Army access to FCI's warehousing infrastructure, modern silos, trained manpower and material-handling equipment during operational contingencies.
  - It will remain valid for **five years,** with an option to extend it by mutual consent.
- **Strategic Significance:** The initiative **strengthens military supply** chains in forward areas, reflecting lessons from **Operation Sindoor** about the importance of resilient logistics during **short-warning or zero-warning** military contingencies.

Read more: **Food Corporation of India**